

HotelAndStars

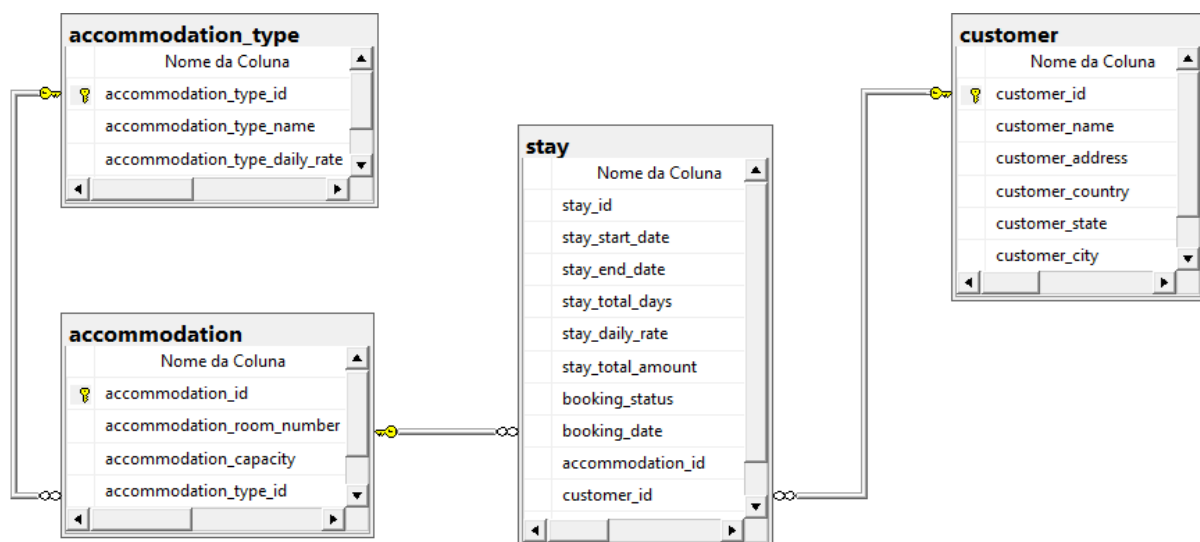
1. Objective.

Build an Entity-Relationship Model (ERM) and, based on it, a Dimensional Model for the manager of the fictional company HotelAndStars, considering the following demands:

- The manager wants to know:
 - How many accommodations and of what type are sold per day;
 - What is the daily revenue;
 - Where the customers are from: country, state, and city;
 - How holidays and school vacations influence accommodations.
- Does not need to know:
 - Who stayed.

2. Entity-Relationship Model (ERM) - SQL Server Management Studio (SSMS).

ERM created:



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3. Dimensional Model - SQL Server Management Studio (SSMS).

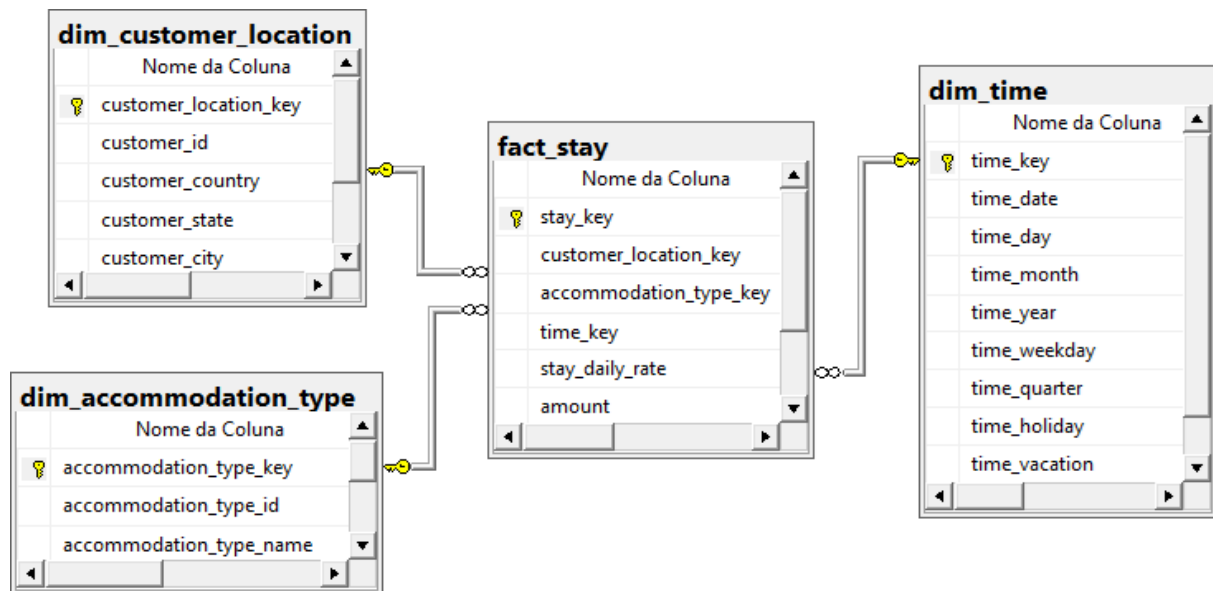
1. To create the analytical database structure model (OLAP) considering the transactional database (OLTP), the subject to be analyzed (the fact) and the dimensions are the various perspectives from which the fact is to be analyzed (the dimensions), using the Star Schema model (the most common model for Business Intelligence), the following transformations were made:

Dimension	Attributes	Source Table
dim_accommodation_type (type of accommodation sold)	accommodation_type_key (SK)	-
	accommodation_type_id	accommodation_type

	accommodation_type_name	accommodation_type
dim_customer_location (where are the customers from?)	customer_location_key (SK)	-
	customer_id	customer
	customer_country	customer
	customer_state	customer
	customer_city	customer
dim_time (dates)	time_key (SK)	-
	time_date	-
	time_day	-
	time_month	-
	time_year	-
	time_weekday	-
	time_quarter	-
	time_holiday	-
	time_vacation	-

Fact	Consolidated table	Granularity	Attributes	Source table
fact_stay	stay	Information on accommodations by day..	stay_key (SK)	-
			customer_location_key (FK)	dim_customer_location
			accommodation_type_key (FK)	dim_accommodation_type
			time_key (FK)	dim_time
			stay_daily_rate	stay
			amount (The default value is 1 for all records, aiming for aggregation by summation)	-

4. Dimensional Model created:



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